

Mars Rovers: Australia's Role in Space Exploration

Name: _____

Date: _____

Year 6 English — Reading

ACARA v9.0: AC9E6LY03 / AC9E6LY05 / AC9E6LY06

Learning Intention

I am learning to analyse how an author uses evidence and text structures to inform and persuade the reader.

Success Criteria

- I can identify the author's purpose and key arguments.
- I can analyse how evidence supports the main ideas.
- I can evaluate the significance of information in the text.

Mars Rovers: Australia's Role in Space Exploration

When people think of space exploration, they often picture NASA control rooms in the United States. But Australia has played a far bigger role in space history than many people realise — a role that continues today with modern Mars exploration missions.

One of Australia's most famous space contributions happened in 1969, during the Apollo 11 Moon landing. As astronauts Neil Armstrong and Buzz Aldrin took their first steps on the Moon, television signals were received and relayed to the world through the Parkes Observatory in New South Wales, home to the giant radio telescope nicknamed "The Dish." For a portion of the broadcast, the historic footage watched by hundreds of millions of people worldwide came directly through Australian technology.

Australia's role in space exploration has only grown since then. The Australian Space Agency, established in 2018, now works with international partners including NASA on numerous missions, including robotic Mars exploration. Australian-developed technology and expertise in areas like remote robotics, communications and mining automation have made Australia a valuable partner for Mars rover missions, since many of the challenges of operating a rover on Mars — extreme distances, harsh terrain, and limited communication — closely resemble challenges faced in Australia's own remote mining and outback regions.

Australian universities and companies have contributed to rover technology, mission communications, and even training scientists who go on to work directly with international Mars missions. The CSIRO's Canberra Deep Space Communication Complex continues to play a critical role today, tracking and communicating with spacecraft — including Mars rovers — as they explore the Martian surface millions of kilometres away.

As space agencies plan future missions to Mars and beyond, Australia's expertise in remote operations, combined with its deep history in space communication, means the country is likely to remain an important, if often overlooked, partner in humanity's exploration of space.

Part A — Comprehension Questions

- Q1. In what year was the Apollo 11 Moon landing footage relayed through Australia? _____
- Q2. What is the nickname of the giant radio telescope at Parkes Observatory? _____
- Q3. True or False: The Australian Space Agency was established before the year 2000. _____
- Q4. Name 2 areas of Australian expertise mentioned that are useful for Mars missions.
1. _____ 2. _____

Mars Rovers: Australia's Role in Space Exploration — Page 2

Name: _____

Date: _____

Year 6 English — Reading

ACARA v9.0: AC9E6LY03 / AC9E6LY05 / AC9E6LY06

Part B — More Comprehension Questions

Q5. Why do the challenges of Mars rover missions resemble challenges in Australia's outback, according to the text?

Q6. What role does the CSIRO's Canberra Deep Space Communication Complex play today?

Q7. What is the author's main purpose in writing this text?

Part C — Challenge: Think Deeper

A student is writing a persuasive speech titled 'Australia deserves more recognition in space history.'

Q8. Identify 2 pieces of evidence from the text the student could use to support this argument.

Q9. Explain why the author chose to describe Australia's role as 'often overlooked' in the final paragraph. What effect does this have on the reader?

Q10. Evaluate how convincing you find the author's argument that Australia is a 'valuable partner' in Mars exploration. Justify your view using evidence and structure from the text.

Answer Key — Part A & B

A1: 1969.

A2: "The Dish."

A3: False — the Australian Space Agency was established in 2018.

A4: Remote robotics and communications (accept mining automation).

A5: Because both involve extreme distances, harsh terrain, and limited communication.

A6: It tracks and communicates with spacecraft, including Mars rovers, as they explore the Martian surface.

A7: To inform readers about Australia's significant but often underappreciated role in space exploration history and modern missions.

Answer Key — Part C

C8: Any 2 reasonable pieces of evidence (e.g. the Parkes Observatory relaying the Moon landing footage, or Australia's expertise in remote robotics being valuable for Mars missions) — teacher to check.

C9: Describing Australia's role as 'often overlooked' draws attention to an unfair gap between Australia's real contribution and public recognition, prompting the reader to feel that Australia deserves more credit — reinforcing the text's persuasive undertone.

C10: Accept a reasoned answer, as long as it references specific evidence (e.g. the Parkes Observatory's historic role, the Australian Space Agency's international partnerships) and text structure (e.g. building from historical to modern examples) to justify the response — teacher to check reasoning quality.